History of Psychology

Session 5: Gestalt and Cognitive Psychology

Rui Mata, Center for Cognitive and Decision Sciences
November 3, 2026

Session information

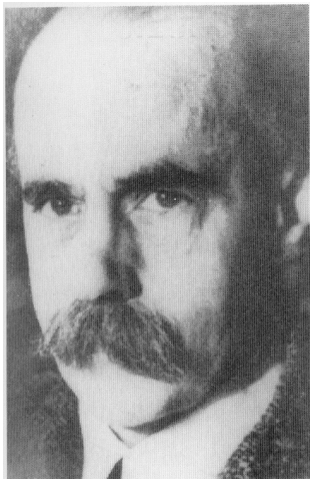
Sessions take place Tuesdays, 8.15-9.45, Kollegienhaus, Aula 033. Slides will be made available shortly before each session.

#	Date	Topic	Instructor	Resources
1	22.09.2026	What kind of science is psychology?	Mata	Spektrum Podcast
2	6.10.2026	The birth of psychology	Mata	Brysbaert & Rastle (2021; 4.1-4.2)
3	13.10.2026	Psychoanalysis	Mata	Brysbaert & Rastle (2021; 4.3-4.4)
4	20.10.2026	Behaviorism	Mata	Brysbaert & Rastle (2021; 5.2)
5	03.11.2026	Gestalt and Cognitive psychology	Mata	Brysbaert & Rastle (2021; 4.2 & 5.3)
6	10.11.2026	Psychology today	Tisdall	Spear (2007)
7	17.11.2026	Psychotherapy research	Tisdall	Braakmann (2015)
8	24.11.2026	Psychological testing	Tisdall	Wasserman (2012)
9	01.12.2026	Decision science	Tisdall	Newell et al. (2022)
10	08.12.2026	What kind of science is psychology? (revisited)	Tisdall	Ball (2012)
11	15.12.2026	Exam (see below)		

Learning Objectives for Today

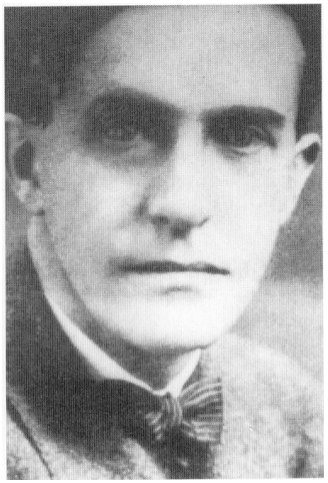
- **Gestalt psychology**: Trace its origins in perception research, identify core concepts (multistability, laws of grouping, and insight), and discuss how these challenged structuralist and behaviourist explanations.
- **Cognitive psychology**: Identify key figures and ideas behind its emergence, discuss tensions between nativist and associationist perspectives, and assess the impact of the “cognitive revolution”.
- Compare schools regarding their core assumptions, methods and explanatory strengths and limitations.

Gestalt Psychology



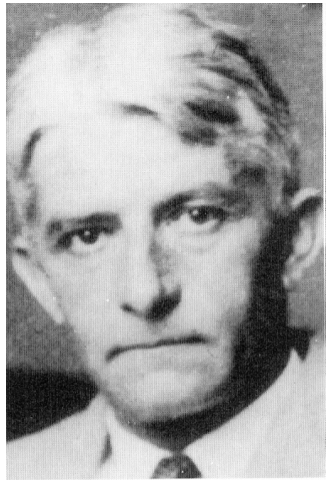
Max Wertheimer (1880-1943)

Wertheimer published a paper *Experimental Studies on the Perception of Movement* (1912) describing the **phi phenomenon** (i.e., the apparent movement caused by alternating light positions). Wertheimer illustrated this phenomenon on an apparatus that utilized two discrete lights on different locations. Although the lights were stationary, flashing the lights at succeeding time intervals (which were systematically varied) caused participants (including Kofka and Köhler!) to perceive the light as moving.



Kurt Koffka (1886-1941)

Koffka exposed american psychologists to Gestalt psychology in 1922 in his article entitled: *Perception: An Introduction to the Gestalt-Theorie*, which appeared in *Psychological Bulletin*.



Wolfgang Köhler (1887-1967)

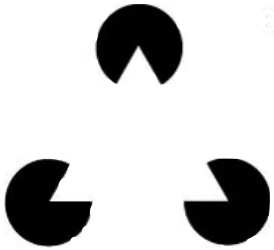
Köhler investigated thinking and reasoning in chimps. He was a critic of both of structuralism and behaviorism.

Gestalt Psychology: Perception

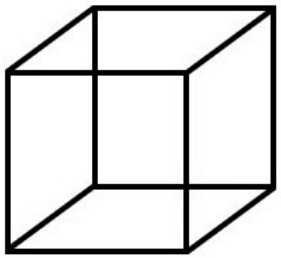
Thesis: The central principle of gestalt psychology is that the mind forms a global whole with self-organizing tendencies. This principle maintains that when the human mind (perceptual system) forms a percept or gestalt, the whole has a reality of its own that goes beyond the individual parts. In this sense, it contrasts with structuralism which aimed to understand specific independent elements of sensation and perception.

"The whole is different than the sum of its parts"

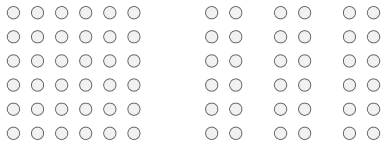
Reification



Multistability



Laws of Grouping



Methods: A combination of experimental manipulation of relevant stimulus factors and observation of behaviour or self-report (introspection).

Vol. 19, No. 10 October, 1922

THE
PSYCHOLOGICAL BULLETIN

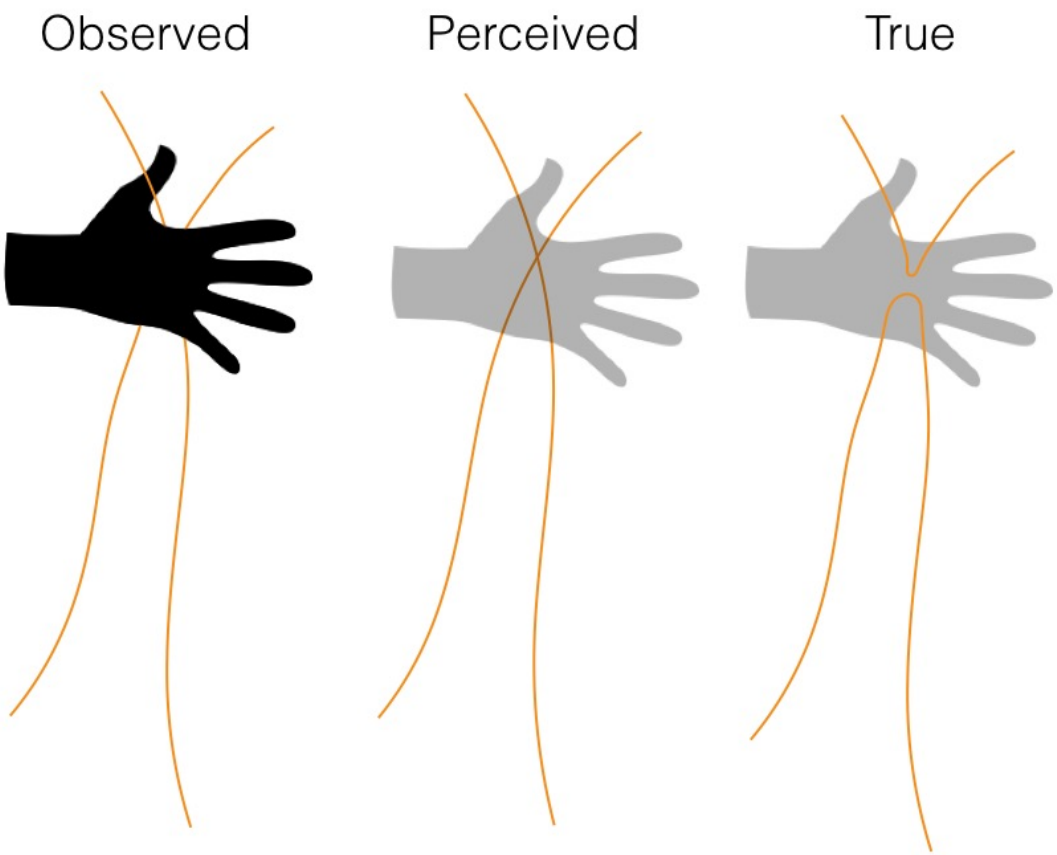
PERCEPTION: AN INTRODUCTION TO THE *GESTALT-THEORIE.*

BY KURT KOFFKA
Giessen

When it was suggested to me that I should write a general critical review of the work recently carried on in the field of perception, I saw an opportunity of introducing to American readers a movement in psychological thought which has developed in Germany during the last ten years. In 1912 Wertheimer stated for the first time the principles of a *Gestalt-Theorie* which has served as the starting point of a small number of German psychologists. Wherever this new method of thinking and working has come in touch with concrete problems, it has not only showed its efficiency, but has also brought to light startling and important facts, which, without the guidance of this theory, could not so easily have been discovered.

The *Gestalt-Theorie* is more than a theory of perception: it is even more than a mere psychological theory. Yet it originated in a study of perception, and the investigation of this topic has furnished the better part of the experimental work which has been done. Consequently, an introduction to this new theory can best be gained, perhaps, by a consideration of the facts of perception.

Gestalt Psychology: Perception (examples)

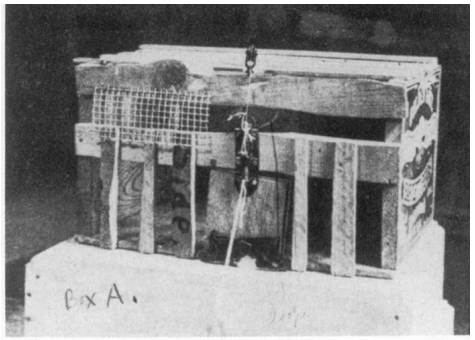


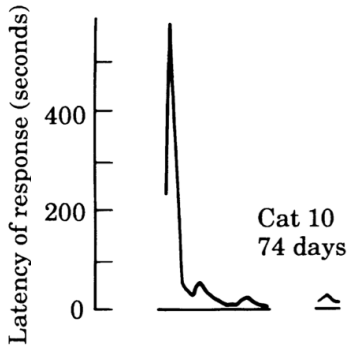
Barnhart, A. S. (2010). The exploitation of Gestalt principles by magicians. Perception, 39(9), 1286–1289. http://doi.org/10.1068/p6766

Gestalt Psychology: Higher-order Thought

Law of effect vs. Insight

Origins of Behaviorism





One of the four puzzle boxes used by Thorndike in the research for his doctoral thesis. From Boakes (1984).

Law of effect

Edward Thorndike, american psychologist conducted research on animal learning using puzzle boxes, that is, apparatuses in which trapped animals could press or pull levers to escape. In 1905, Thorndike proposed the law of effect that states that "responses that produce a satisfying effect in a particular situation become more likely to occur again in that situation, and responses that produce a discomforting effect become less likely to occur again in that situation."

https://en.wikipedia.org/wiki/Law_of_effect



Insight

Wolfgang Köhler was director of the Prussian Academy of Sciences anthropoid research station, Tenerife, Canary Islands, where he wrote a book on problem solving titled *The Mentality of Apes* (1917). In this research, Köhler observed how chimpanzees solve problems, such as that of retrieving bananas when positioned out of reach by using, for example, stacked wooden crates as makeshift ladders. Köhler concluded that the chimps had not arrived at these methods through trial-and-error (which Thorndike had claimed to be the basis of all animal learning), but rather that they had an insight, in which, having realized the answer, they proceeded to carry it out in a purposeful manner.

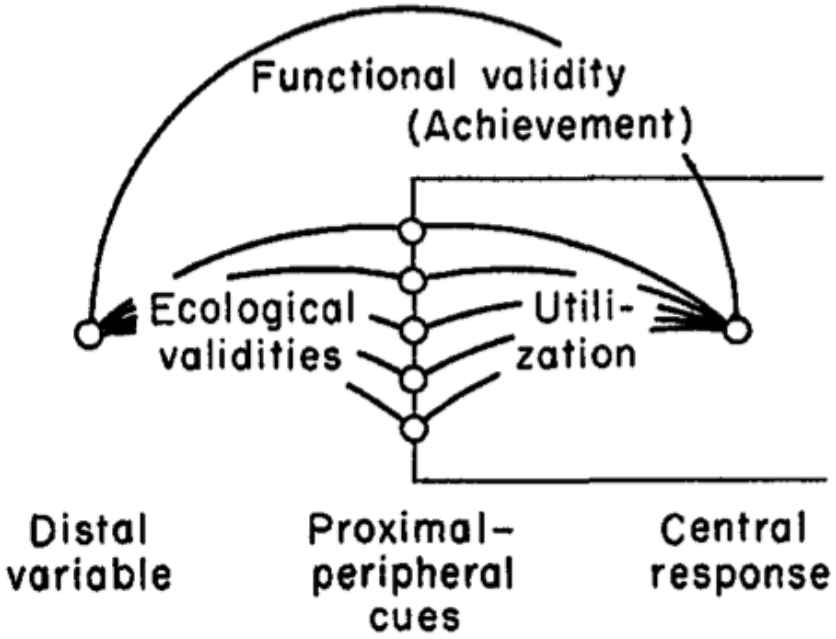
https://en.wikipedia.org/wiki/Insight

Gestalt Psychology: Impact



Egon Brunswick
(1903-1955)

Brunswick proposed the **lens model** which describes the link between characteristics of the world and individual's perception of these characteristics.



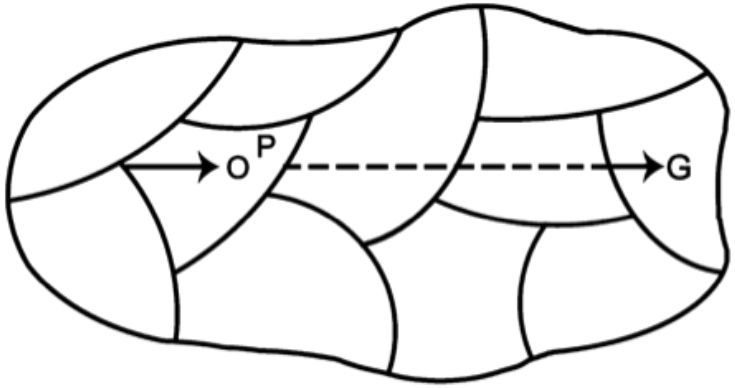
Egon Brunswik (born in Budapest, studied in Vienna, later emigrated to USA) argued that psychology should give as much attention to the properties of the organism's environment as it does to the organism itself. He asserted that the environment with which the organism comes into contact is an uncertain, probabilistic one, however lawful it may be in terms of physical principles. Adaptation to a probabilistic world requires that the organism learn to employ probabilistic uncertain evidence (proximal cues) about the world (the distal object). His work has influenced psychology of perception (cf. Roger Shepard) and judgment and decision making (cf. Ken Hammond). His focus on the environment also led him to argue for the need to use representative designs in psychology (i.e., naturalistic sampling of stimuli)

Gestalt Psychology: Impact



Kurt Lewin
(1890-1947)

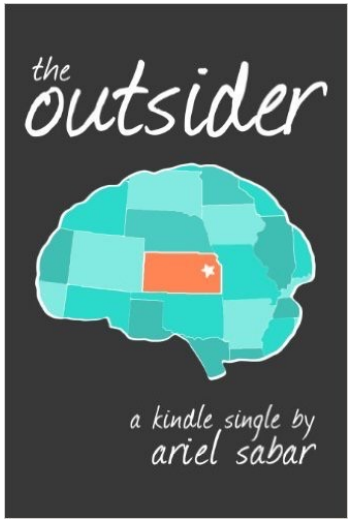
Lewin proposed **field theory**, which states that a person's life is made up of multiple distinct spaces, which the person has to travail in order to achieve his/her goals.



Kurt Lewin (born in Germany but emigrated to the United States) was a psychologist, known as one of the modern pioneers of social, organizational, and applied psychology. Lewin proposed that human behavior is always a function of the person in their environment and which he summarised in a simple equation:

$B = f(P, E)$

Levin's work influenced and encouraged many (social) psychologists to consider how behaviour is embedded in social context (cf. Roger Barker's biography and his goal of studying the "naturally occurring behavior of free-ranging persons").



Gestalt Psychology: Impact

Gestalt psychologists don't make it to the top 10…

Historians' and Chairpersons' Rankings of the Importance of All-Time and Contemporary Psychologists

Historians[a]			Chairpersons[b]		
Rank	Individual	Rank points	Rank	Individual	Rank points
		All-time rankings			
1	Wundt, W.	189	1	Skinner, B. F.	508
2	James, W.	167	2	Freud, S.	459
3	Freud, S.	156	3	James, W.	372
4	Watson, J.	108	4	Piaget, J.	237
5	Pavlov, I.	79	5	Hall, G. S.	216
6	Ebbinghaus, H.	69	6	Wundt, W.	203
7	Piaget, J.	51	7	Rogers, C.	192
8	Skinner, B. F.	46	8	Watson, J.	188
9	Binet, A.	46	9	Pavlov, I.	152
10	Fechner, G.	46	10	Thorndike, E. L.	124

Korn, J. H., Davis, R., & Davis, S. F. (1991). Historians' and chairpersons' judgments of eminence among psychologists. *American Psychologist, 46*(7), 789–792. doi: 10.1037/0003-066X.46.7.789

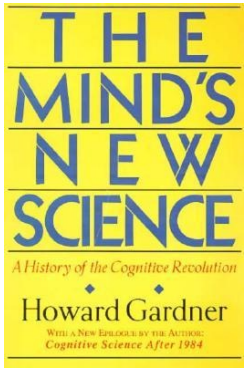
Gestalt Psychology: Impact

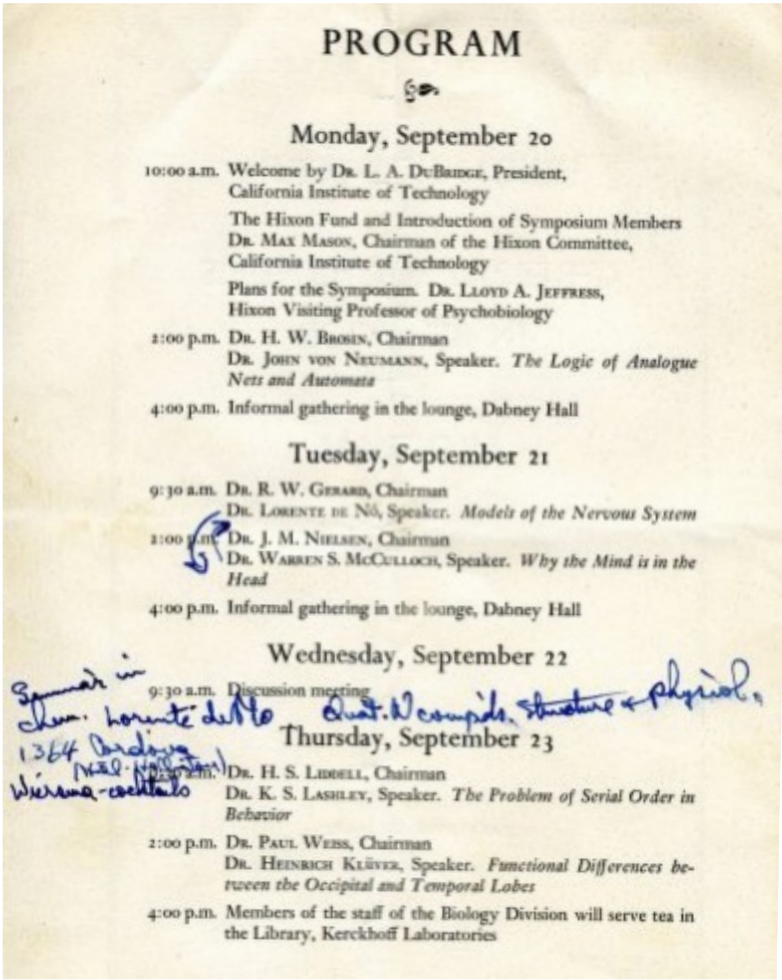
"The Behaviorist's premises, we remember, lead to certain expectations and experiments. What I have just said invites us to proceed in another direction. I suggest that, in this situation, we forget about schools. The Behaviorist is convinced that his functional concepts are those which we all ought to use. The Gestalt psychologist, who deals with a greater variety of both phenomenal and physical concepts, expects more from work based on such premises. Both parties feel that their procedures are scientifically sound. Why should we fight? Many experiments done by Behaviorists seem to me to be very good experiments. May I now ask the Behaviorists to regard the use of some phenomenal facts, and also of field physics, as perfectly permissible? If we were to agree on these points, we could, I am sure, do excellent work together. It would be an extraordinary experience -- and good for psychology."

Köhler, W. (1959). Gestalt psychology today. American Psychologist, 14, 727-734.
http://psychclassics.yorku.ca/Kohler/today.htm

Cognitive Psychology: The Hixon Symposium (1948)

The **1948** Hixon Symposium marks the beginning of a period in which scientists, including mathematicians and neurophysiologists, proposed a parallel between mind and machine (i.e., information-processing logical devices) and strongly criticised behavioural theories…





PROGRAM

Monday, September 20

10:00 a.m. Welcome by Dr. L. A. DuBridge, President, California Institute of Technology

The Hixon Fund and Introduction of Symposium Members Dr. Max Mason, Chairman of the Hixon Committee, California Institute of Technology

Plans for the Symposium. Dr. Lloyd A. Jeffress, Hixon Visiting Professor of Psychobiology

2:00 p.m. Dr. H. W. Brosin, Chairman
Dr. John von Neumann, Speaker. *The Logic of Analogue Nets and Automata*

4:00 p.m. Informal gathering in the lounge, Dabney Hall

Tuesday, September 21

9:30 a.m. Dr. R. W. Gerard, Chairman
Dr. Lorente de Nó, Speaker. *Models of the Nervous System*

2:00 p.m. Dr. J. M. Nielsen, Chairman
Dr. Warren S. McCulloch, Speaker. *Why the Mind is in the Head*

4:00 p.m. Informal gathering in the lounge, Dabney Hall

Wednesday, September 22

9:30 a.m. Discussion meeting

Thursday, September 23

[illegible] Dr. H. S. Liddell, Chairman
Dr. K. S. Lashley, Speaker. *The Problem of Serial Order in Behavior*

2:00 p.m. Dr. Paul Weiss, Chairman
Dr. Heinrich Klüver, Speaker. *Functional Differences between the Occipital and Temporal Lobes*

4:00 p.m. Members of the staff of the Biology Division will serve tea in the Library, Kerckhoff Laboratories

Hixon Symposium on Cerebral Mechanisms in Behavior, California Institute of Technology, Pacific State Hospital, Pomona, California. September 20, 1948

Gardner, H. (1985). The mind's new science: A history of the cognitive revolution. New York: Basic Books.

Cognitive Psychology: The Hixon Symposium (1948)



Karl Lashley
(1890-1958)

Zoologist/Psychologist, worked with John Watson on learning in rats, conducted studies to observe the effects of cortical lesions on trained rats. In 1951, Lashley published a famous paper called “The Problem of Serial Order in Behavior,” in which he pointed out that complex sequential behavior (such as playing a piece on the piano) could not be executed by one response sending a proprioceptive signal back to the brain which would then trigger the next response in the sequence – as there wasn’t enough time for the neural signals to travel up to the brain and back down. Instead, behavior had to be controlled by a central, hierarchically organized program.



Warren McCulloch
(1898-1969)

Neurophysiologist, made contributions to cybernetics. For example, wrote with Walter Pitts "A Logical Calculus of the Ideas Immanent in Nervous Activity” (1943) to demonstrate that a Turing machine program could be implemented in a finite network of formal neurons, that the neuron was the base logic unit of the brain. Later, Pitts and McCulloch suggested how "nervous nets" can be used to recognise visual inputs despite changes in orientation or size.



John von Neumann
(1903-1957)

Mathematician, physicist, computer scientist, made major contributions to a number of fields, including foundations of mathematics, computing of self-replicating machines (cells), and game theory.

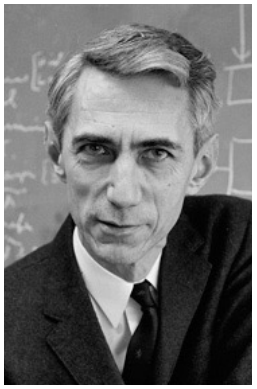
Cognitive Psychology: The Dartmouth Workshop (1956)

In 1956, a summer workshop initiated by John McCarthy (University of Dartmouth, USA) considered by many the seminal event for artificial intelligence, convened key figures in the artificial intelligence and cognitive psychology movement (the names below are just some of those who attended).

Solomonoff, G.. Ray Solomonoff and the Dartmouth Summer Research Project in Artificial Intelligence, 1956. Retrieved from http://raysolomonoff.com/dartmouth/dartray.pdf

Cognitive Psychology: The Dartmouth Workshop (1956)

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Claude Shannon
(1916-2001)

information theory: defined information mathematically, how information can be communicated over (noisy) limited channels, and how it can be reconstructed with low probability of error

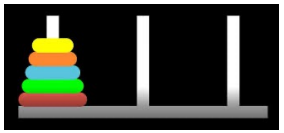


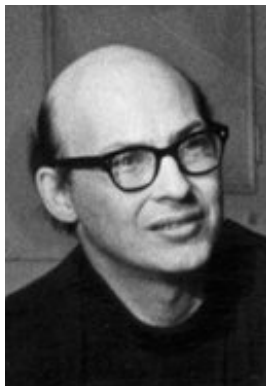
Allen Newell
(1927-1992)



Herbert Simon
(1916-2001)

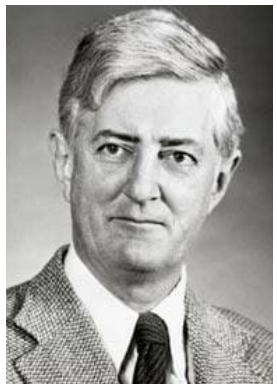
general problem solver: computer program that could solve simple problems (e.g., Tower of Hanoi)





Marvin Minsky
(1927-2016)

artificial neural networks: computing systems inspired by the biological neural networks that constitute animal brains



Oliver Selfridge
(1926-2008)

Solomonoff, G.. Ray Solomonoff and the Dartmouth Summer Research Project in Artificial Intelligence, 1956. Retrieved from http://raysolomonoff.com/dartmouth/dartray.pdf

Noam Chomsky: (innate) Universal Grammar



https://www.youtube.com/watch?v=MLk47AMBdTA

Noam Chomsky: (innate) Universal Grammar



4

A Review of B. F. Skinner's *Verbal Behavior*

Noam Chomsky

1

A great many linguists and philosophers concerned with language have expressed the hope that their studies might ultimately be embedded in a framework provided by behaviorist psychology, and that refractory areas of investigation, particularly those in which meaning is involved, will in this way be opened up to fruitful exploration. Since this volume [*Verbal Behavior* (New York: Appleton-Century-Crofts, 1957)—*Ed.*] is the first large-scale attempt to incorporate the major aspects of linguistic behavior within a behaviorist framework, it merits and will undoubtedly receive careful attention. Skinner is noted for his contributions to the study of animal behavior. The book under review is the product of study of linguistic behavior extending over more than twenty years. Earlier versions of it have been fairly widely circulated, and there are quite a few references in the psychological literature to its major ideas.

The problem to which this book is addressed is that of giving a "functional analysis" of verbal behavior. By functional analysis, Skinner means identification of the variables that control this behavior and specification of how they interact to determine a particular verbal response. Furthermore, the controlling variables are to be described completely in terms of such notions as *stimulus, reinforcement, deprivation,* which have been given a reasonably clear meaning in animal experimentation. In other words, the goal of the book is to provide a way to predict and control verbal behavior by observing and manipulating the physical environment of the speaker.

Skinner feels that recent advances in the laboratory study of animal behavior permit us to approach this problem with a certain optimism, since "the basic processes and relations which give verbal behavior its special characteristics are now fairly well understood . . . the results [of this experimental work] have been surprisingly free of species restrictions. Recent work has shown that the methods can be extended to human behavior without serious modification (3).[1]

It is important to see clearly just what it is in Skinner's program and claims that makes them appear so bold and remark-

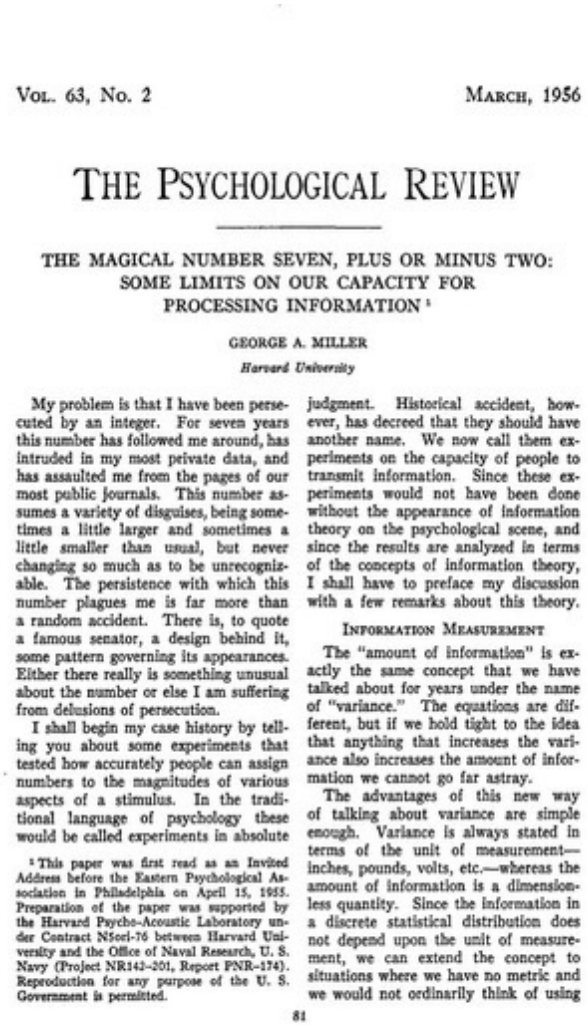
From *Language* 35, no. 1 (1959): 26-58. Reprinted by permission of the Linguistic Society of America and the author. Sections 5-10 have been omitted (the notes are therefore not numbered consecutively).

1959

"Chomsky has been one of behaviorism's most successful and damaging critics. In a review of Skinner's book on verbal behavior, **Chomsky (1959)** charged that behaviorist models of language learning cannot explain various facts about language acquisition, such as the rapid acquisition of language by young children, which is sometimes referred to as the phenomenon of "lexical explosion." A child's linguistic abilities appear to be radically underdetermined by the evidence of verbal behavior offered to the child in the short period in which he or she expresses those abilities. By the age of four or five (normal) children have an almost limitless capacity to understand and produce sentences which they have never heard before. Chomsky also argued that it seems plainly untrue that language learning depends on the application of reinforcement. A child does not, as an English speaker in the presence of a house, utter "house" repeatedly in the presence of reinforcing elders. Language as such seems to be learned without, in a sense, being explicitly taught or taught in detail, and behaviorism doesn't offer an account of how this could be so. (…) Our history of reinforcement is often too impoverished to determine uniquely what we do or how we do it. Much learning, therefore, seems to require pre-existing or innate representational structures or principled constraints within which learning occurs."

Graham, G. (2017), "Behaviorism", The Stanford Encyclopedia of Philosophy. Edward N. Zalta (ed.). Retrieved from https://plato.stanford.edu/archives/spr2017/entries/behaviorism/

The Harvard Center for Cognitive Studies



VOL. 63, NO. 2 MARCH, 1956

THE PSYCHOLOGICAL REVIEW

THE MAGICAL NUMBER SEVEN, PLUS OR MINUS TWO: SOME LIMITS ON OUR CAPACITY FOR PROCESSING INFORMATION[1]

GEORGE A. MILLER

Harvard University

My problem is that I have been persecuted by an integer. For seven years this number has followed me around, has intruded in my most private data, and has assaulted me from the pages of our most public journals. This number assumes a variety of disguises, being sometimes a little larger and sometimes a little smaller than usual, but never changing so much as to be unrecognizable. The persistence with which this number plagues me is far more than a random accident. There is, to quote a famous senator, a design behind it, some pattern governing its appearances. Either there really is something unusual about the number or else I am suffering from delusions of persecution.

I shall begin my case history by telling you about some experiments that tested how accurately people can assign numbers to the magnitudes of various aspects of a stimulus. In the traditional language of psychology these would be called experiments in absolute judgment. Historical accident, however, has decreed that they should have another name. We now call them experiments on the capacity of people to transmit information. Since these experiments would not have been done without the appearance of information theory on the psychological scene, and since the results are analyzed in terms of the concepts of information theory, I shall have to preface my discussion with a few remarks about this theory.

INFORMATION MEASUREMENT

The "amount of information" is exactly the same concept that we have talked about for years under the name of "variance." The equations are different, but if we hold tight to the idea that anything that increases the variance also increases the amount of information we cannot go far astray.

The advantages of this new way of talking about variance are simple enough. Variance is always stated in terms of the unit of measurement—inches, pounds, volts, etc.—whereas the amount of information is a dimensionless quantity. Since the information in a discrete statistical distribution does not depend upon the unit of measurement, we can extend the concept to situations where we have no metric and we would not ordinarily think of using

[1] This paper was first read as an Invited Address before the Eastern Psychological Association in Philadelphia on April 15, 1955. Preparation of the paper was supported by the Harvard Psycho-Acoustic Laboratory under Contract N5ori-76 between Harvard University and the Office of Naval Research, U. S. Navy (Project NR142-201, Report PNR-174). Reproduction for any purpose of the U. S. Government is permitted.

81

1956



George Miller
(1920-2012)

Miller observed that memory span of young adults is approximately seven items and noticed that memory span is approximately the same for stimuli with vastly different amount of information (e.g., letters, digits, words). Miller concluded that memory span is not limited in terms of bits but rather in terms of chunks. More importantly, his paper became an important manifesto for the idea that the mind could be studied and described like an information processing system with significant capacity limitations.

Miller, G. (1956). The magical number seven, plus or minus two: Some limits on our capacity for processing information. *Psychological Review, 63*, 81-97.

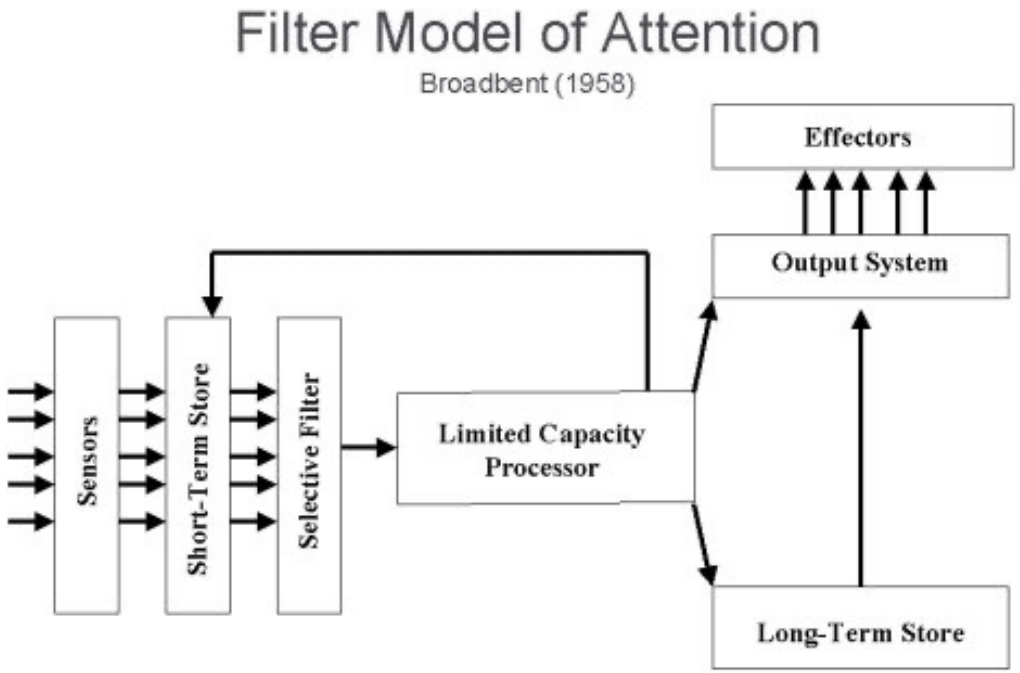
The Harvard Center for Cognitive Studies

"Whatever we called it, **the cognitive counter-revolution in psychology brought the mind back into experimental psychology**. I think it is important to remember that the mind had never disappeared from social or clinical psychology. It was only experimentalists in the US who really believed that behaviorism would work. In my own case, when I became dissatisfied at Harvard between B.F. Skinner 's strict behaviorism and S.S. Stevens' psychophysics, I turned to Jerry Bruner's social psychology, and in 1960 that led to the creation at Harvard of the Center for Cognitive Studies. Bruner's group at Bow Street had been calling themselves the 'Cognition Project' for some time, so we simply changed it from a project to a center. (…) Behaviorism flourished primarily in the US and this cognitive revolution in psychology re-opened communication with some distinguished psychologists abroad. In Cambridge, UK, Sir Frederic Bartlett's work on memory and thinking had remained unaffected by behaviorism. In Geneva, Jean Piaget's insights into the minds of children had inspired a small army of followers. And in Moscow, A.R. Luria was one of the first to see the brain and mind as a whole. None of these three spent time at the Center but we knew their work well. Whenever we doubted ourselves we thought of such people and took courage from their accomplishments. (…) The bright young graduates grew up to become important psychologists unafraid of words like ***mind*** and ***expectation*** and ***perception*** and ***memory***.

Miller, G. (2003). The cognitive revolution: A historical perspective. ***Trends in Cognitive Sciences, 7***(3), 141-144.

Cognitive Psychology: Information Processing

Psychology became fascinated with information-processing diagrams, in which boxes and arrows stand for (temporary) stores of information and/or processes that transform the information and the transmission of information.

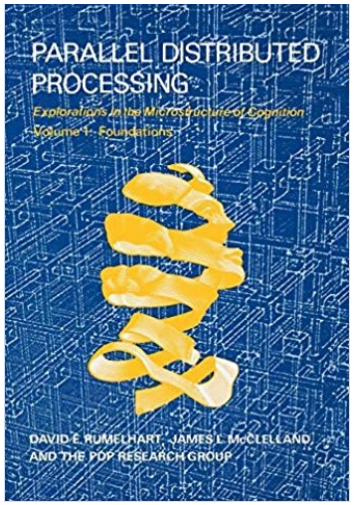


Broadbent, D. (1958). ***Perception and Communication***. London: Pergamon Press.

Cognitive Psychology: Connectionism

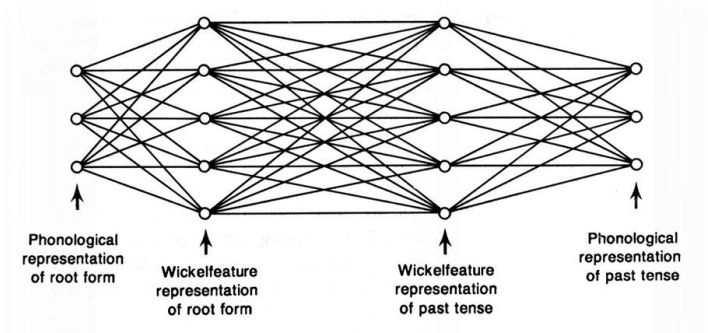


David Rumelhart James McClelland



1986

"Connectionism is a movement in cognitive science that hopes to explain intellectual abilities using artificial neural networks (also known as "neural networks" or "neural nets"). Neural networks are simplified models of the brain composed of large numbers of units (the analogs of neurons) together with weights that measure the strength of connections between the units. These weights model the effects of the synapses that link one neuron to another. Experiments on models of this kind have demonstrated an ability to learn such skills as face recognition, reading, and the detection of simple grammatical structure."



Connectionism (parallel distributed processing) presented itself as a plausible, mechanistic alternative to innate (grammatical) principles….

Buckner, C. & Garson, J. (2019), "Connectionism", The Stanford Encyclopedia of Philosophy. Edward N. Zalta (ed.). Retrieved from https://plato.stanford.edu/archives/spr2017/entries/connectionism/

Summary

- **Gestalt psychology:** perception depends on organised wholes rather than isolated elements ("the whole differs from the sum of its parts"). Methods: experiments and observational studies identifying laws of perceptual organisation; Köhler extended the approach to learning and reasoning through insight. Contribution and limits: challenged trial-and-error learning as a general explanation of cognition and highlighted context (Brunswik, Lewin), yet offered few explicit models of underlying mechanisms.

- **Cognitive psychology:** behaviour depends on mental representations and information processing, often understood through the computer analogy. Methods: experiments on attention, language and memory, alongside computational models using symbolic rules or connectionist networks. Contribution and limits: Lashley and Chomsky challenged behaviourist explanations; cognitive psychology developed explicit models of mental processes, while giving comparatively little attention to affect and, at least initially, questions about neural implementation.

WHAT DO YOU THINK?

Compare and contrast the schools: behaviourism, gestalt, cognitive psychology (structuralism, functionalism, psychoanalysis)

